

1. Introduction

The study focuses on improving the estimation of the **autocovariance sequence** for **reversible Markov chains**. Accurate estimation of the autocovariance sequence is critical in various applications, particularly in **Markov Chain Monte Carlo (MCMC)** simulations, where it helps in the estimation of **asymptotic variance**. The asymptotic variance quantifies the long-term variability in estimates, playing a crucial role in assessing the reliability of MCMC simulations.

Problem Statement: Traditional methods like **batch means (BM)** and **spectral variance estimators (SVE)** have limitations, especially in providing consistent asymptotic variance estimation. These methods often require careful tuning of parameters (e.g., batch size, window size), making them less robust or less accurate under certain conditions. The goal of this study is to introduce a **shape-constrained estimator**, called the **Moment Least Squares Estimator (Moment LSE)**, that addresses these limitations by leveraging the moment sequence representation of the autocovariance sequence.

Motivation: Improving the estimation of the autocovariance sequence can lead to better performance in MCMC simulations, where precise variance estimates are critical for drawing reliable conclusions in applications ranging from scientific research to financial modeling.

2. Key Contributions

The study introduces the **Moment Least Squares Estimator (Moment LSE)**, which improves on traditional methods by imposing shape constraints derived from the moment sequence structure of the autocovariance sequence. The key contributions of the study are:

1. **Shape-Constrained Estimator:** The estimator uses the **moment sequence representation** of the autocovariance sequence to ensure that the estimated autocovariances follow the correct structural patterns.
 2. **Theoretical Guarantees:** The estimator is shown to be **strongly consistent** in the ℓ_2 -norm for geometrically ergodic reversible Markov chains. This means that the estimator gets more accurate as the sample size increases, which is a critical property for variance estimation in MCMC.
 3. **Improved Empirical Performance:** The Moment LSE outperforms state-of-the-art estimators like batch means and spectral variance estimators in simulations, demonstrating better accuracy and consistency.
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3. Background: Autocovariance Sequence

The **autocovariance sequence** of a Markov chain helps quantify how much the current state of the chain is related to its past states. For a reversible Markov chain, the autocovariance sequence is defined as:

$$\gamma(k) = E_{\pi}[(g(X_0) - \mu)(g(X_k) - \mu)]$$

where $\gamma(k)$ measures how much the value at time $t + k$ is related to the value at time t , and μ is the mean of the process. This sequence captures the memory of the Markov chain—how much influence past values have on the present.

In **MCMC**, the **asymptotic variance** σ^2 , which reflects the long-run uncertainty of estimates, is related

$$\sigma^2 = \sum_{k=-\infty}^{\infty} \gamma(k)$$

to the sum of the autocovariances:

Accurate estimation of the autocovariance sequence is crucial for computing reliable estimates of the asymptotic variance.

4. Approach: Moment Least Squares Estimator (Moment LSE)

The Moment LSE is designed to improve the estimation of the autocovariance sequence by incorporating **shape constraints** derived from the moment sequence representation of the autocovariances. The estimator works as follows:

1. **Initial Empirical Estimate:** The process starts by calculating an initial empirical estimate of the autocovariance sequence, denoted $r_M(k)$, from the MCMC samples. This raw estimate is typically noisy and may not respect the true structure of the autocovariance sequence.
2. **ℓ_2 -Norm Projection:** To refine the estimate, the Moment LSE projects the initial empirical estimate onto the space of sequences that satisfy the moment sequence representation. This projection minimizes the ℓ_2 -norm (squared error) between the initial estimate and the final shape-constrained sequence:

$$\hat{m} = \operatorname{argmin}_{m \in M_{\infty}(\delta)} \sum_k (r_M(k) - m(k))^2$$

where $M_{\infty}(\delta)$ represents the space of sequences that admit a moment sequence representation.

3. **Shape Constraints:** The projection ensures that the final estimate respects **monotonicity** and **convexity** constraints, which are natural properties of the autocovariance sequence in reversible Markov chains.
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5. Comparison with Traditional Methods

The study compares the performance of the Moment LSE with several traditional methods, including **Batch Means (BM)**, **Overlapping Batch Means (OLBM)**, and **Initial Sequence Estimators (Init.cnv, Init.dec, Init.pos)**. Each method has its strengths and limitations:

- **Batch Means (BM):** Simple to compute but sensitive to batch size. If the batch size is too small, the variance estimate can be biased.
- **Spectral Variance Estimator (SVE):** Consistent but requires tuning of the window size, which can be tricky and lead to suboptimal results.
- **Initial Convex, Initial Decreasing, Initial Positive:** These methods offer simplicity but lack strong theoretical guarantees and may not perform as well in all cases.

The **Moment LSE** outperforms these traditional methods by providing more accurate and consistent estimates of the autocovariance sequence, as it avoids the need for tuning parameters and is backed by strong consistency results.

6. Empirical Results

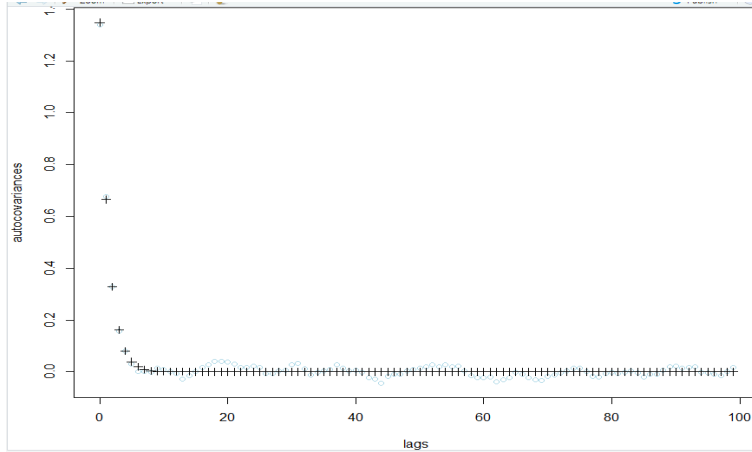
Performance Evaluation:

In the study, the Moment LSE was tested on simulated data, particularly an **AR(1) chain** with $\rho = 0.5$ and a chain length of 10,000. The results were compared with other estimators, and the key performance metric was **Mean Squared Error (MSE)**.

Estimator	Average_MSE	SE
BM	0.304	0.051
Init.conv	0.069	0.012
Init.dec	0.082	0.014
Init.pos	0.103	0.018
MomentLS	0.054	0.012
MomentLS.oracle	0.024	0.004
OLBM	0.206	0.030

Key Observations:

- **Moment LSE** achieves the lowest MSE compared to other estimators, demonstrating superior accuracy in estimating the autocovariance sequence.
- The **oracle version** of Moment LSE (using the known value of ρ) achieves even lower MSE, illustrating the potential of the method when optimal parameters are available.
- Traditional methods like **Batch Means** and **OLBM** perform worse due to their sensitivity to batch or window size tuning.
- **Autocovariance Comparison:**



The study also compared the empirical autocovariances with the estimates from Moment LSE. The Moment LSE was able to closely match the true autocovariances, maintaining consistency across different lags. This result further supports the claim that the shape constraints imposed by Moment LSE lead to more accurate estimates.

7. Conclusion

The **Moment Least Squares Estimator (Moment LSE)** provides a novel and effective method for estimating the autocovariance sequence in reversible Markov chains. By leveraging the moment sequence representation and imposing shape constraints, the Moment LSE achieves strong theoretical guarantees, such as ℓ_2 -norm consistency, and outperforms traditional estimators like Batch Means and Spectral Variance Estimators in empirical studies.

Key Contributions:

- The introduction of a shape-constrained estimator that addresses the limitations of existing methods.
- Demonstrated strong empirical performance, with lower MSE and better accuracy in variance estimation.

- Theoretical guarantees of consistency for geometrically ergodic reversible Markov chains, providing confidence in the long-term reliability of the estimator.

In conclusion, the Moment LSE is a valuable tool for MCMC variance estimation and has the potential for broader application in time series analysis and other fields where autocovariance estimation is critical.